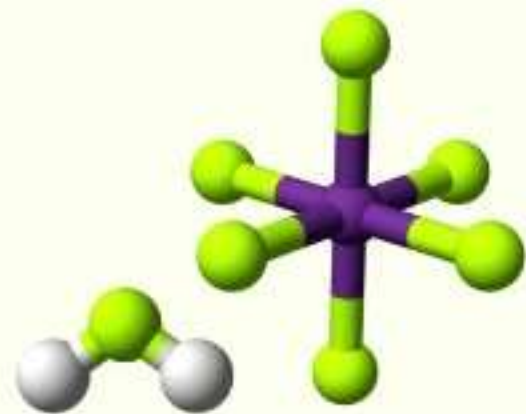
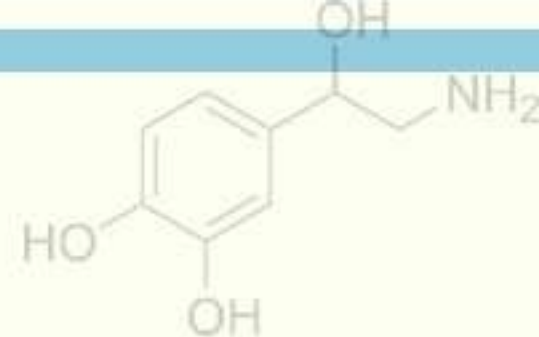


CHEMISTRY

THE WORLD OF MATTER - 2

MODULE - 1





CLASSIFICATION OF IMPURE MATTER

Impure Matter

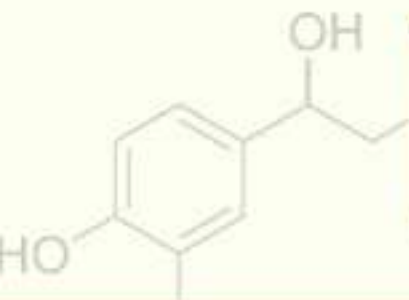
Homogeneous mixture

solution

Heterogeneous mixture

suspension

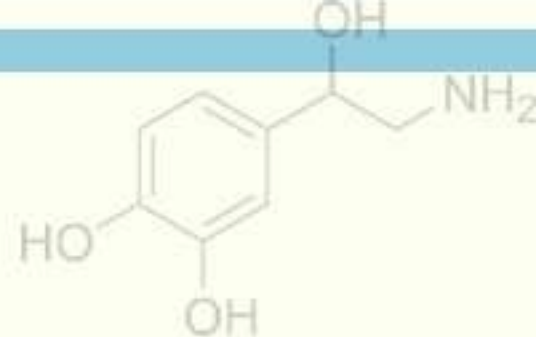
colloid



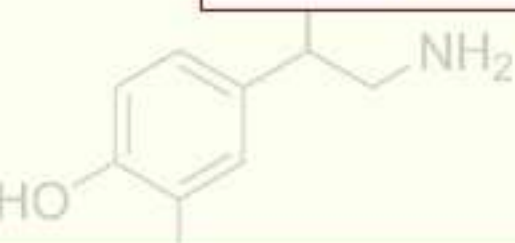


IMPURE MATTER

HOMOGENOUS MIXTURE

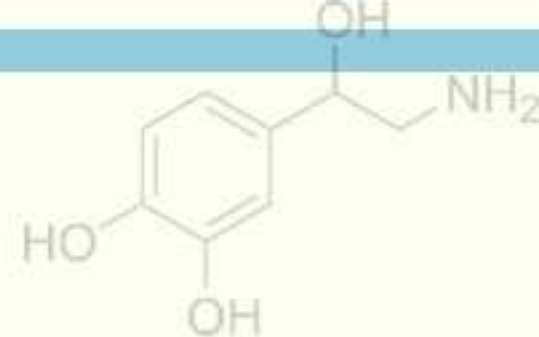


- 1 A mixture which has uniform properties throughout.
Ex. Salt dissolves in water, it mixes with the entire volume of water.
- 2 Homogenous mixtures are called SOLUTIONS
- 3 Solutions have two components **Solute and Solvent**.
Solute is the small amount of impurity that gets dissolved in some larger quantities called solvent.





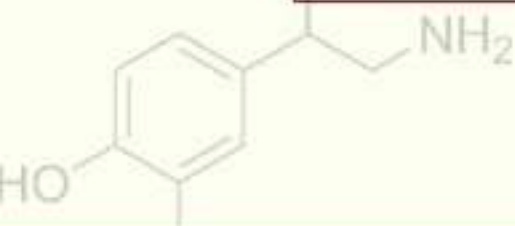
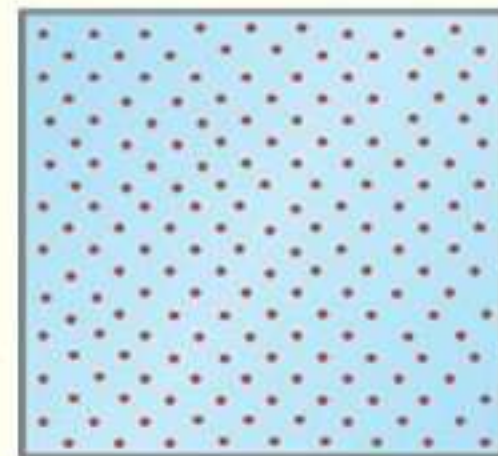
PROPERTIES OF SOLUTION

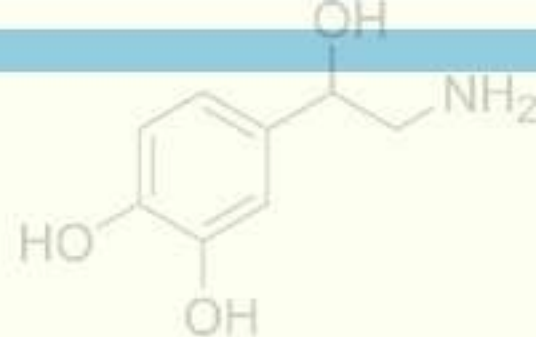


1 A solution is a homogeneous mixture. Thus the particles are **uniformly distributed** in the solvent

2 The solute particles are **smaller than 1 nanometer in diameter**. So **they don't scatter light** passing through them.

3 A solution is stable that means the solute particles do not settle down the bottom of the container when left undisturbed.

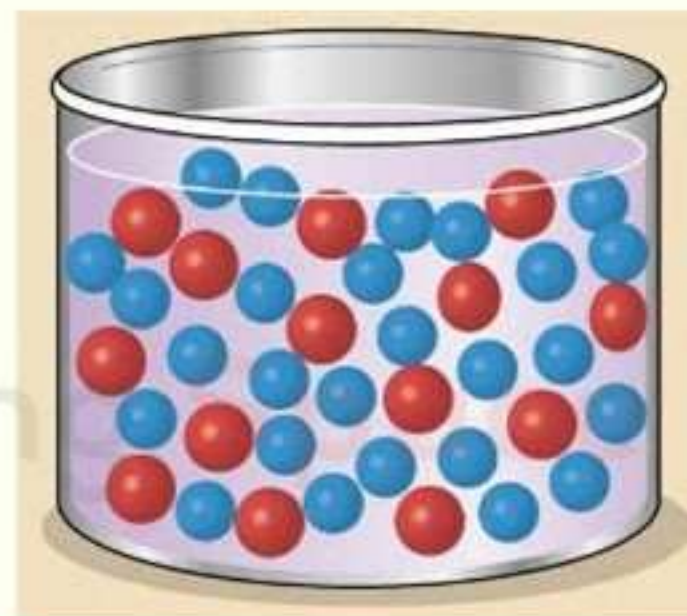




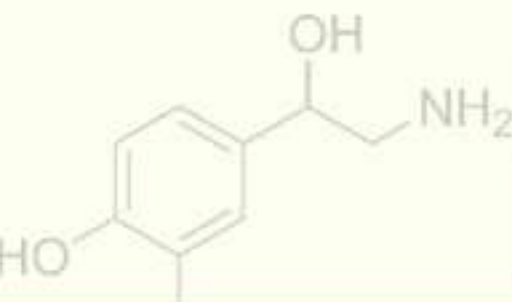
SOLUTE

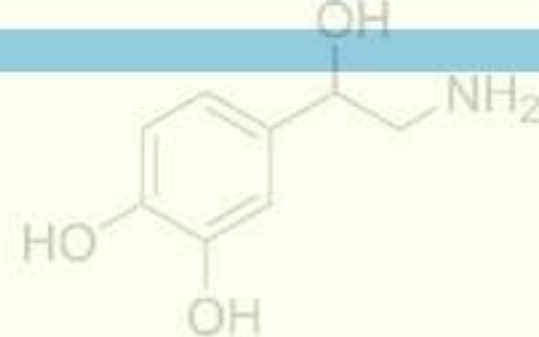


SOLVENT

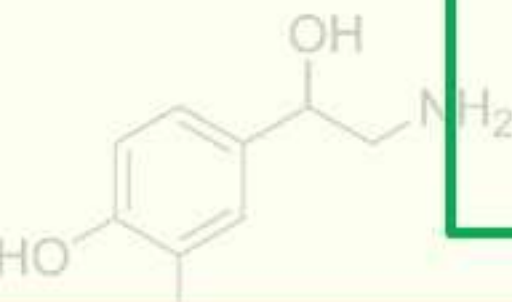
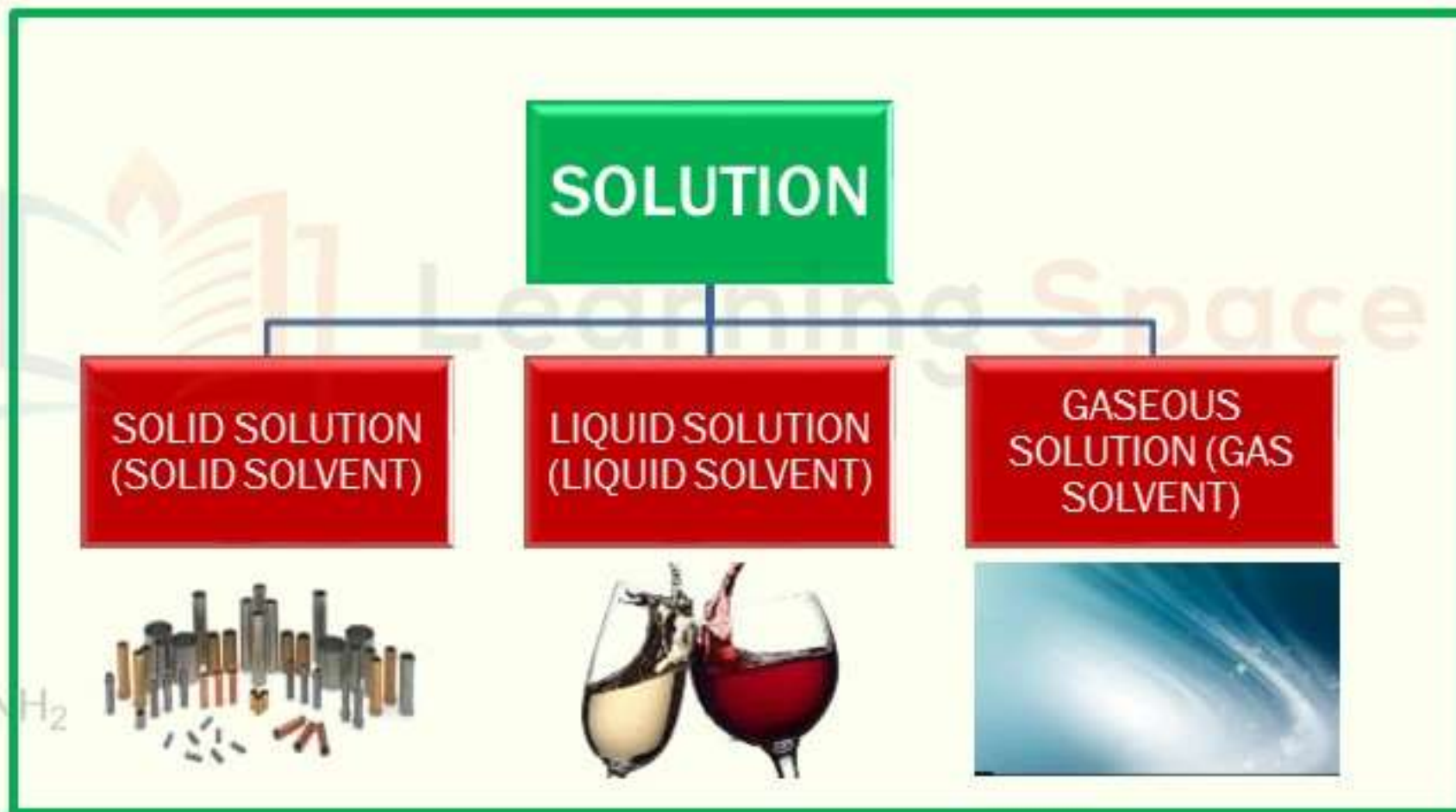


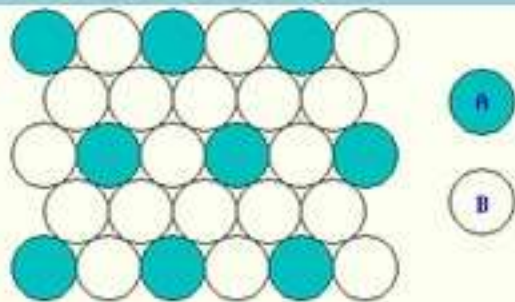
SOLUTION = SOLUTE + SOLVENT



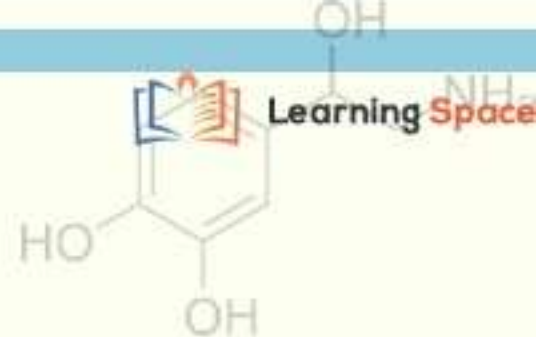


TYPES OF SOLUTION





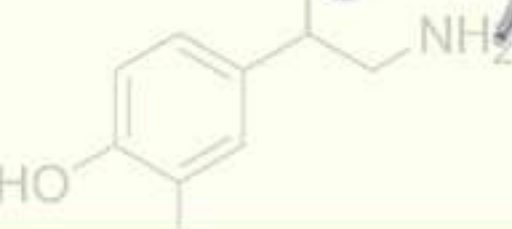
THE SOLID SOLUTIONS

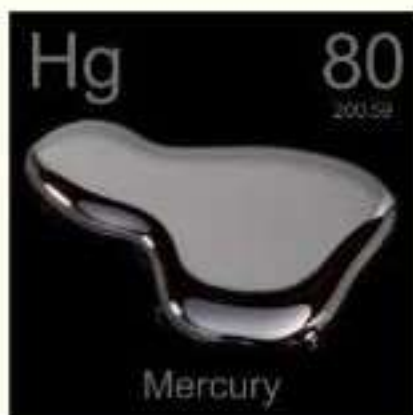


1

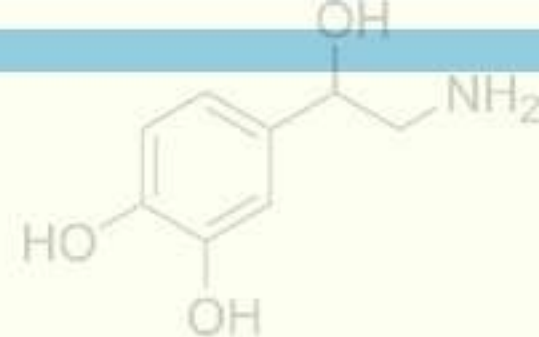
STAINLESS STEEL

- Made by mixing different liquefied metals.
- The main components are Iron Carbon, nickel and chromium.
- When a metal is dissolved in liquid form with another metal and non metal and re-solidified it creates an **ALLOY**.
- Thus alloy is a **solid in solid solution**.





THE SOLID SOLUTIONS



2

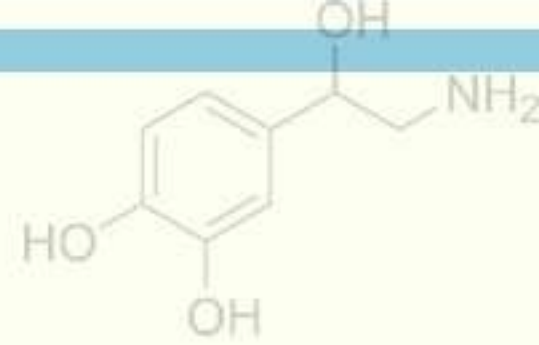
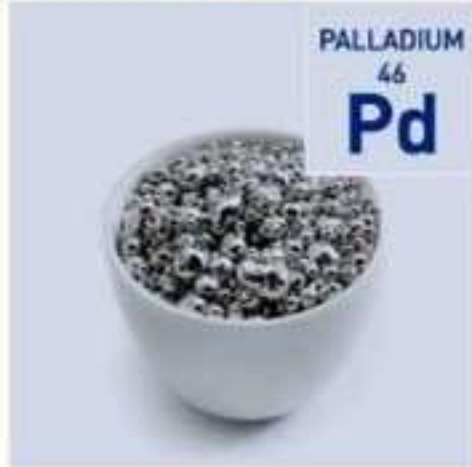
AMALGAM

- Silver Amalgam is widely used in dental fillings.
- Amalgam contains mercury which is the only liquid metal at room temperature.
- The amalgam is a solution of liquid mercury in a solid metal.
- Amalgam is a **liquid in solid solution**.





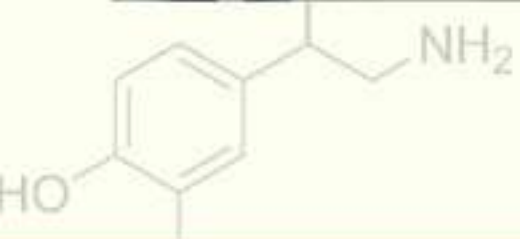
THE SOLID SOLUTIONS



3

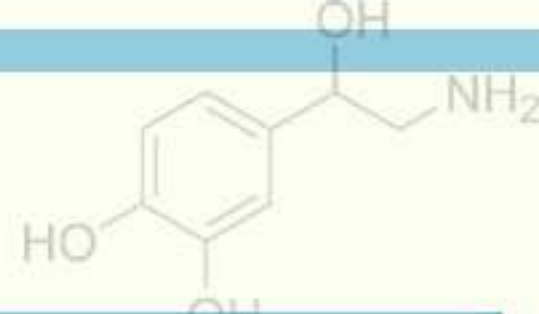
HYDROGEN FUEL

- Hydrogen is a new fuel for cars.
- To store pure hydrogen gas palladium metals are used.
- Hydrogen gas can be readily adsorbed (absorbed on the surface) on palladium.
- This is a **gas in solid solution**.



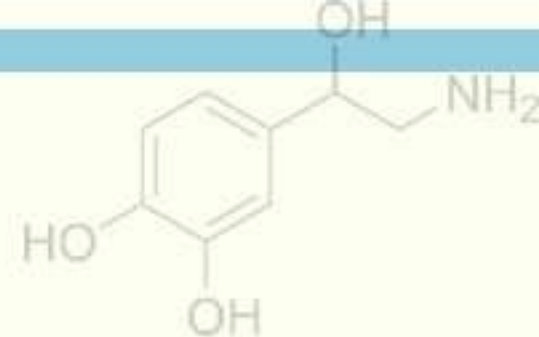


THE SOLID SOLUTIONS



Solvent	solute	Solution	EXAMPLES
Solid	Solid	Alloy E.g., Stainless steel made up with iron carbon nickel cobalt Other e.g. Bronze and Brass	
Solid	Liquid	Amalgam E.g., mercury in silver mercury in sodium Mercury in gold	
Solid	Gas	Adsorption of gases on metals E.g., adsorption of hydrogen on palladium	





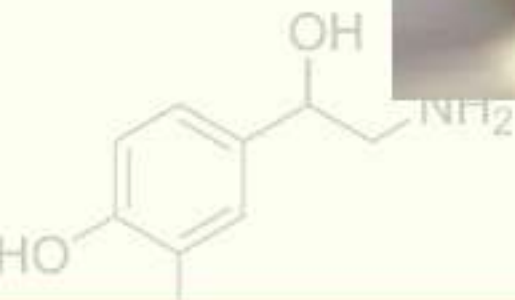
THE LIQUID SOLUTIONS

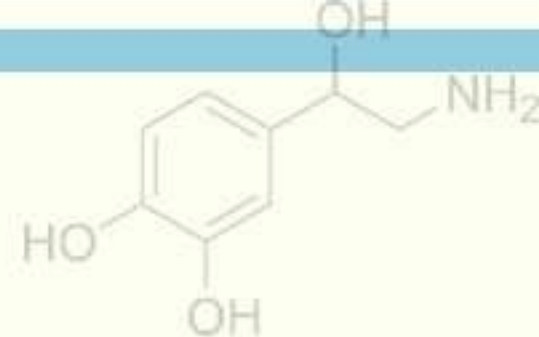
1

SUGAR SYRUP



- Sugar syrup is a sweet liquid mixture.
- It is made by mixing the sugar crystals in water
- Solid sugar is a solute mixed in the solvent water
- This is a **solid in liquid solution**





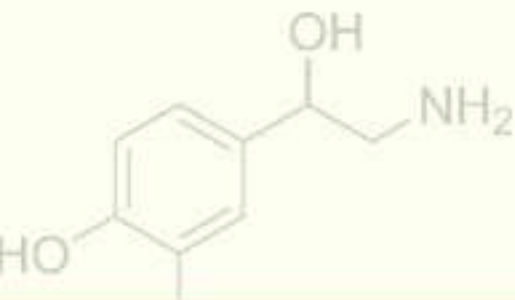
THE LIQUID SOLUTIONS

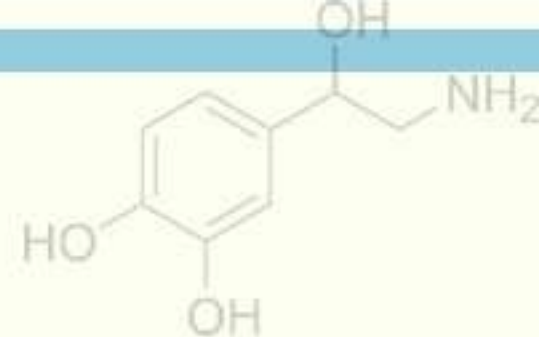


2

VINEGAR

- Vinegar is made up of acetic acid
- Acetic acid is dissolved in water to make it edible
- Acetic acid is solute in solvent water
- This is a liquid in liquid solution





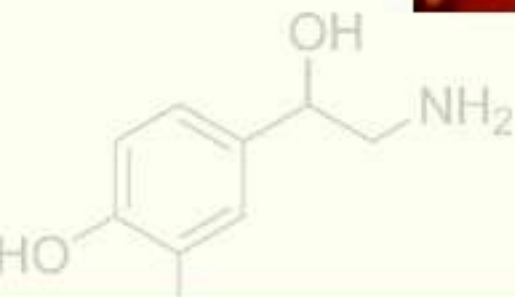
THE LIQUID SOLUTIONS

3

AERATED DRINK

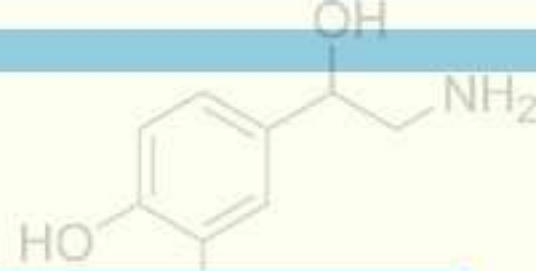


- Aerated drink is mixed with gas like CO_2 .
- This gas gives the fizz and the flavor
- Gas is solute in liquid solvent
- This is a **gas in liquid solution**.

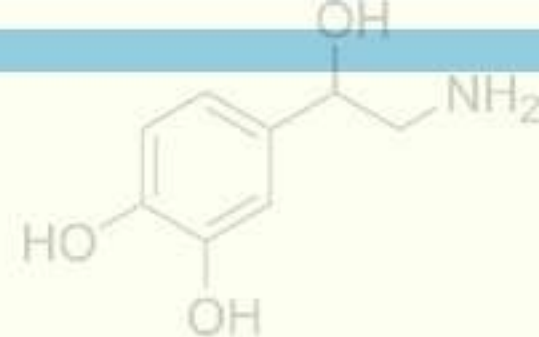




THE LIQUID SOLUTIONS



Solvent	Solute	Solution	EXAMPLES
Liquid	solid	Sugar solution	
liquid	liquid	Vinegar Other e.g. all alcoholic beverages	
liquid	gas	Soda water /Aerated drink	



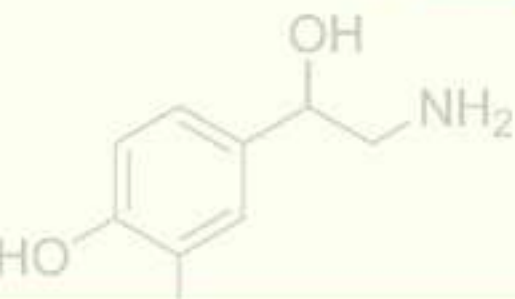
THE GAS SOLUTIONS

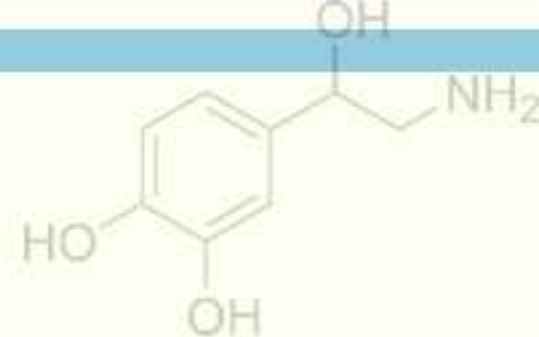
1

CAMPHOR



- Camphor is used in Aarti or worship etc.
- Camphor can directly sublime into air
- Camphor particles mixes with gaseous air
- This is a **solid in gas solution**.

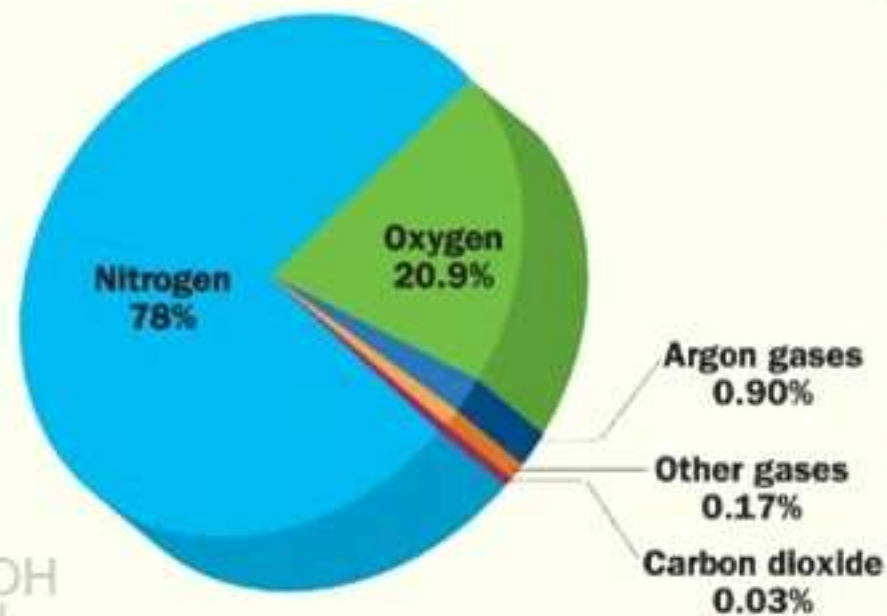




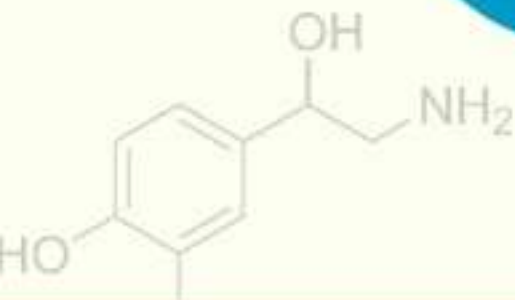
THE GAS SOLUTIONS

2

AIR AROUND US

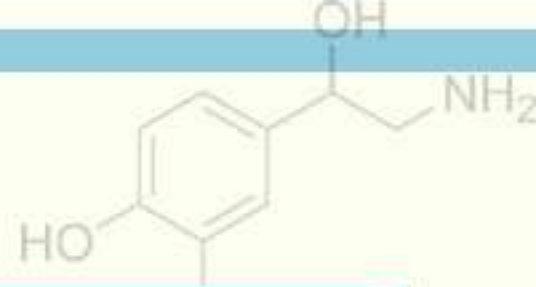


- The air we breathe is also a mixture.
- Different gases like Nitrogen Oxygen and carbon dioxide etc. are mixed together
- Gas is mixed with another gases uniformly
- This is a **gas in gas solution**.

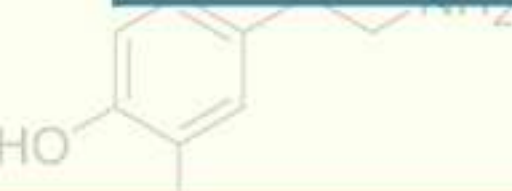




THE GASEOUS SOLUTIONS

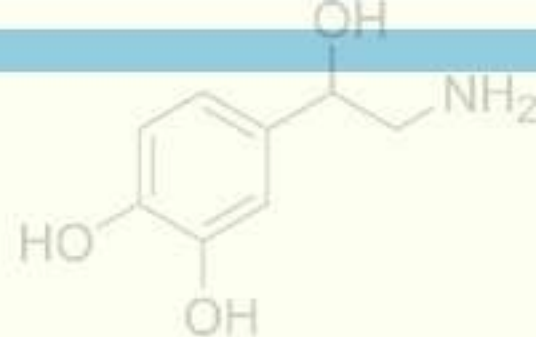


Solvent	Solute	Solution	EXAMPLES
Gas	Solid	Camphor vapour in air	
Gas	Gas	Atmospheric Air - its different components like Nitrogen Oxygen and Carbon dioxide	

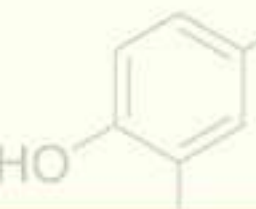


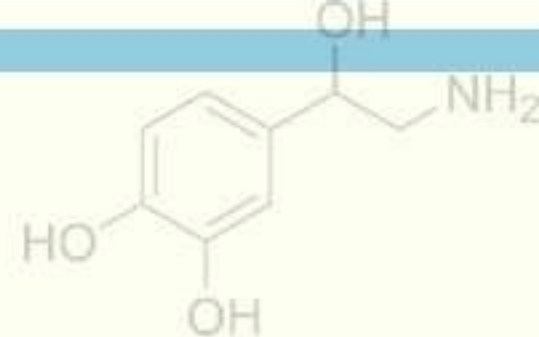


SOLUBILITY IN SOLUTIONS



- Solubility is the property of a solid, liquid, or gaseous substance called solute to dissolve in a solid, liquid or gaseous solvent.
- Solubility is **temperature dependent**. Solubility generally increases with increasing temperature.
- It is due to that we feel hot food less salty than cold ones or while making sugar syrup we can add large quantity of sugar while heating the mixture.





SOLUBILITY IN SOLUTIONS- STAGES OF SOLUTION

1. UNSATURATED



Solute is less than the quantity that can be dissolved

2. SATURATED

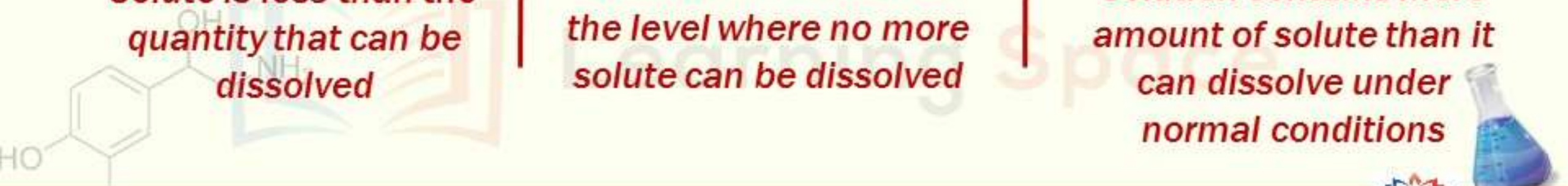


Solution has reached the level where no more solute can be dissolved

3. SUPERSATURATED

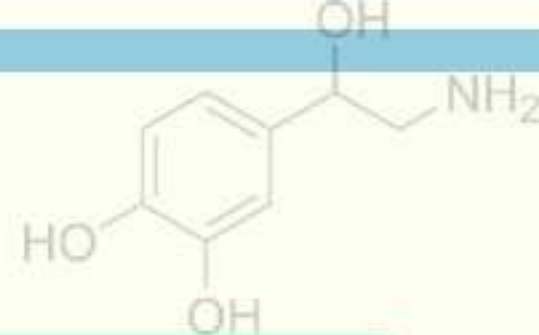


Solution contains more amount of solute than it can dissolve under normal conditions





CLASSIFICATION OF IMPURE MATTER



Impure Matter

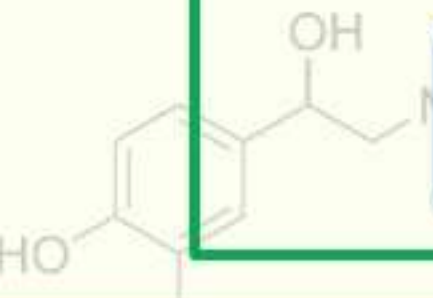
Homogeneous mixture

solution

Heterogeneous mixture

suspension

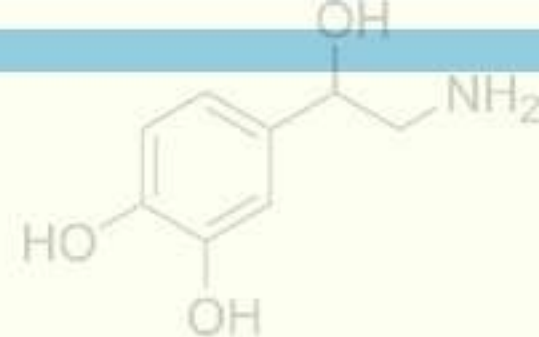
colloid





IMPURE MATTER

HETEROGENEOUS MIXTURE



HETEROGENEOUS MIXTURES ARE NON UNIFORM IN COMPOSITION.
THEY ARE NON-HOMOGENEOUS SYSTEMS

1

Non Homogeneous systems in which the solids are dispersed in liquids are called **Suspensions**. Particles of suspension are visible to the naked eye like sand in water.



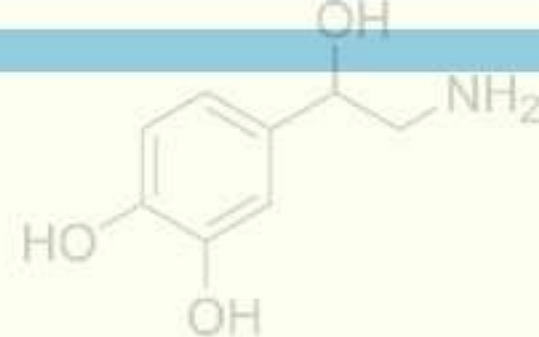
2

The heterogeneous mixture in which the particles are uniformly mixed in the medium and not visible through the naked eye is called **Colloids**.





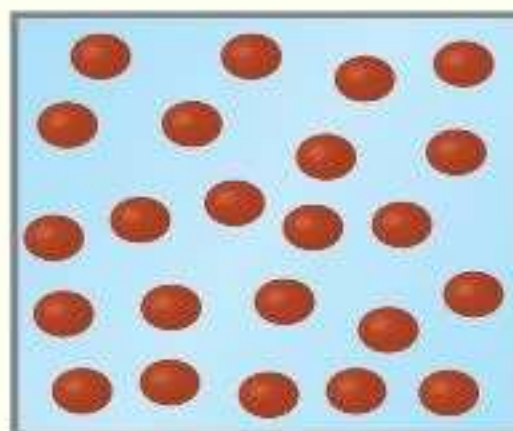
COMPARE THE PARTICLE SIZE IN ALL TYPES OF IMPURE MATTER



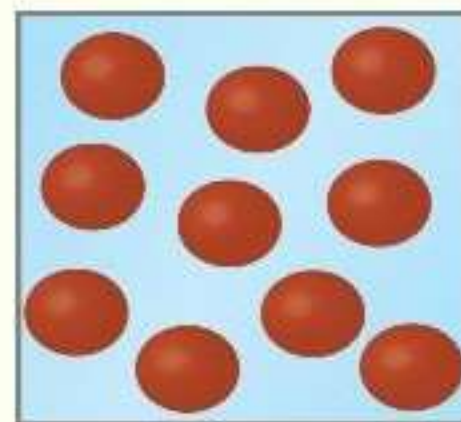
SOLUTION



COLLOID



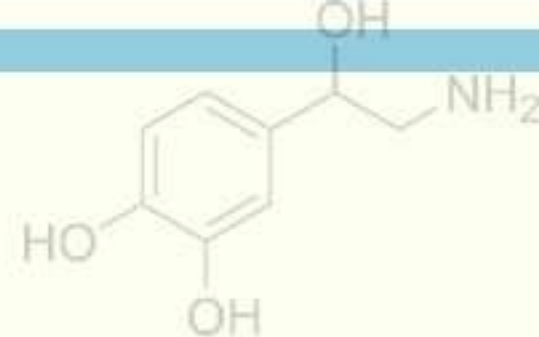
SUSPENSION



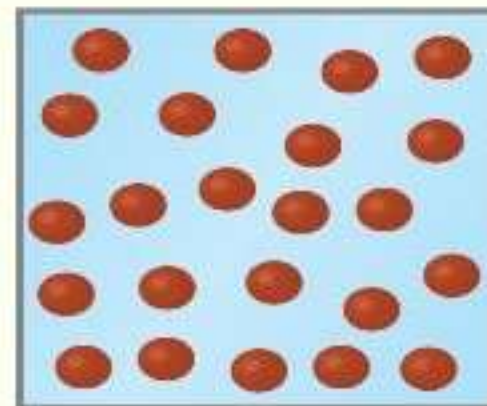


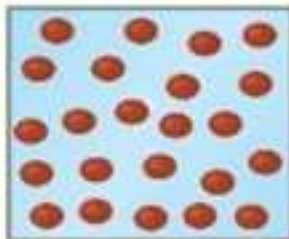
HETEROGENOUS MIXTURE

COLLOIDS

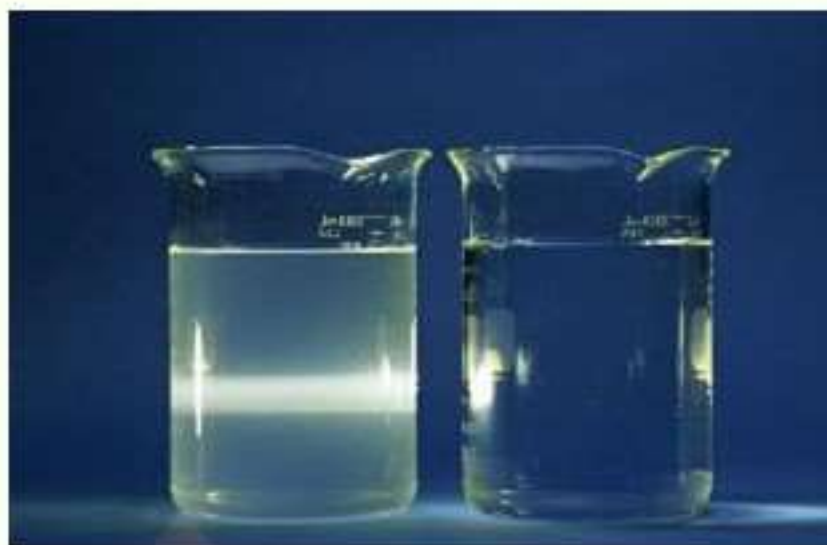
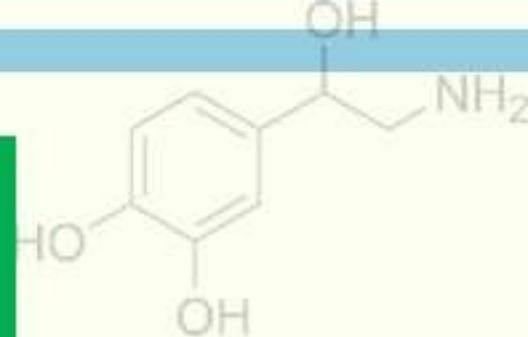


- 1 Colloids are heterogeneous mixture in which particles are uniformly suspended in another.
- 2 Colloidal particles are big enough to not get dissolved and are small enough to remain uniformly in the solvent.
- 3 Particles of colloid are small so they cannot be separated by regular filtration techniques unlike suspended particles.





THE SCATTERING OF LIGHT COLLOIDS - TYNDALL EFFECT

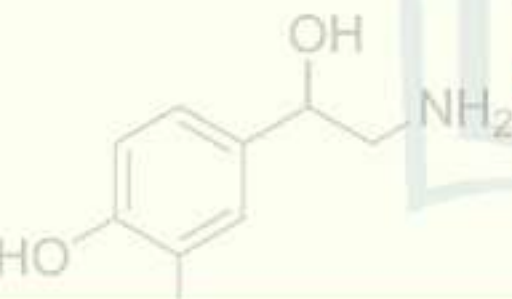


SEEN IN DETTOL SOLUTION



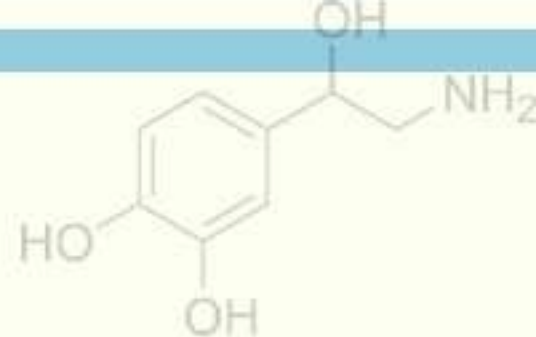
SEEN IN OUR ATMOSPHERE

Colloidal particles have a particular dimension that they can scatter light. So, the path of light passing through a colloidal mixture is always visible





WHAT DOES A COLLOID COMPRISE OF?

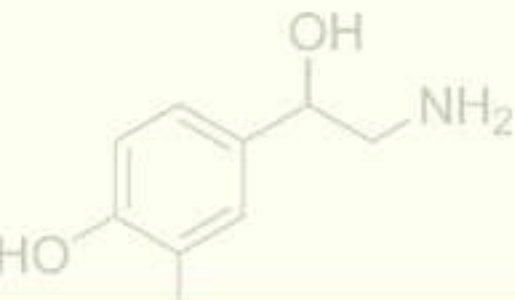
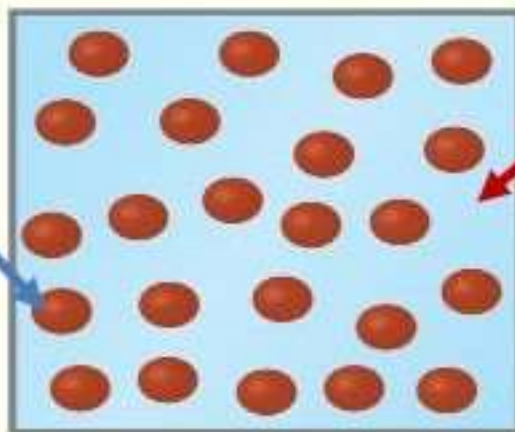


1. DISPERSED PHASE

The finely divided particles which are present in a continuous system. They are present in a two phased system

2. DISPERSED MEDIUM

A continuous medium in which the dispersed phase (solid liquid, gaseous) is distributed over



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